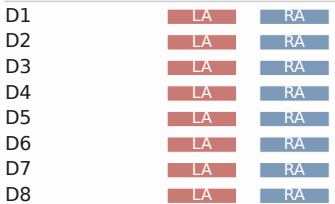
a Healthy left-right chamber asymmetry defined on donor-paired samples

Input: healthy paired atria

8 donor pairs; each donor contributes one LA and one RA sample



Primitive unit: paired contrast ($LA_d - RA_d$) within each donor.

Naive baseline

$y \sim \text{chamber}$

drops donor pairing
weaker chamber prior

Chamber-aware primitive definition

Donor-level equations used to build the healthy chamber prior

$$(1)\quad \mathrm{aCAI}_{d,p} = \frac{1}{|G_p|}\sum_{g \in G_p} E^{LA}_{g,d} - \frac{1}{|G_p|}\sum_{g \in G_p} E^{RA}_{g,d}$$

$$(2)\quad \bar{\mathrm{aCAI}}_p = \frac{1}{D}\sum_{d=1}^{D} \mathrm{aCAI}_{d,p}$$

$$(3)\quad c_p = \frac{1}{D}\sum_{d=1}^{D} \mathbf{1}\left[\mathrm{sign}(\mathrm{aCAI}_{d,p}) = \mathrm{sign}(\bar{\mathrm{aCAI}}_p)\right]$$

$D = 8$ donors; G_p genes in program p; E normalized expression.
Sign rule: $\bar{\mathrm{aCAI}}_p > 0$ (LA-biased), $\bar{\mathrm{aCAI}}_p < 0$ (RA-biased).

LA = red, RA = blue; Figure 1A uses healthy paired donors only (no rhythm split).